

AAROH — Sample Learning Material

Mathematics — Class 4

Topic: Fractions and Basic Operations

Learning Objectives

- Understand fractions as parts of a whole.
- Identify numerator and denominator.
- Compare simple fractions.
- Add and subtract fractions with the same denominator.
- Apply fractions to simple real-life problems.

1. Understanding Fractions

A fraction represents a part of a whole. If a whole object is divided into 4 equal parts and one part is taken, the fraction is $\frac{1}{4}$. The top number is called the numerator. It tells us how many parts we have. The bottom number is called the denominator. It tells us how many equal parts make the whole.

2. Examples

Fraction	Numerator	Denominator	Meaning
$\frac{1}{2}$	1	2	1 out of 2 equal parts
$\frac{3}{4}$	3	4	3 out of 4 equal parts
$\frac{2}{5}$	2	5	2 out of 5 equal parts
$\frac{5}{8}$	5	8	5 out of 8 equal parts

3. Comparing Fractions

When two fractions have the same denominator, the fraction with the larger numerator is larger. For example, $\frac{3}{7}$ is greater than $\frac{2}{7}$ because 3 parts are more than 2 parts when the whole is divided into seven equal parts.

4. Adding Fractions with the Same Denominator

When fractions have the same denominator, add their numerators and keep the denominator the same. For example: $\frac{2}{7} + \frac{3}{7} = \frac{5}{7}$.

5. Subtracting Fractions with the Same Denominator

When fractions have the same denominator, subtract the smaller numerator from the larger numerator and keep the denominator the same. For example: $\frac{6}{8} - \frac{2}{8} = \frac{4}{8}$.

6. Real-Life Example

Rina has a chocolate bar divided into 8 equal pieces. She eats 3 pieces. What fraction of the chocolate bar did she eat? What fraction is left?

Practice Questions

1. What is the numerator of $\frac{5}{9}$?
2. What is the denominator of $\frac{7}{10}$?
3. Which is greater: $\frac{3}{6}$ or $\frac{5}{6}$?
4. Solve: $\frac{2}{9} + \frac{4}{9}$.
5. Solve: $\frac{7}{8} - \frac{3}{8}$.
6. A pizza is divided into 8 equal pieces. Aman eats 3 pieces. What fraction did he eat?
7. Create one real-life example of a fraction.

Key Concepts

Fraction: A part of a whole. **Numerator:** The top number of a fraction. **Denominator:** The bottom number of a fraction. **Equal parts:** Parts of the same size.

Testing purpose: Upload this PDF to AAROH to test content extraction, concept detection, English → Santali/Hindi translation, simplification, quiz generation, assignment generation, flashcards and RAG using mathematics content.